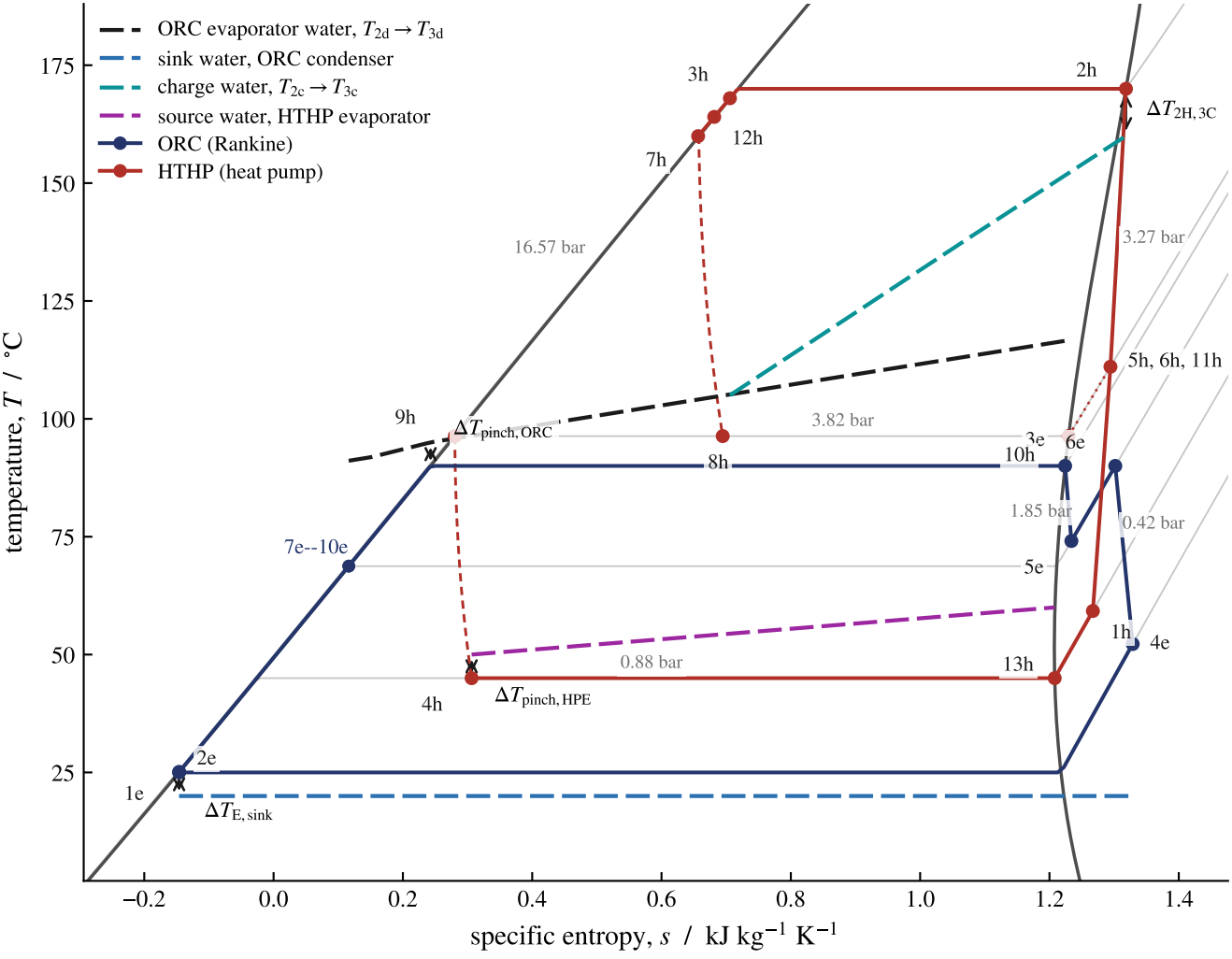
ORC evaporator water, $T_{2d} \rightarrow T_{3d}$
sink water, ORC condenser
charge water, $T_{2c} \rightarrow T_{3c}$
source water, HTHP evaporator
ORC (Rankine)
HTHP (heat pump)
temperature, T / °C
specific entropy, s / kJ kg^{-1} K^{-1}
175
150
125
100
75
50
25
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
3h
2h
12h
7h
16.57 bar
$\Delta T_{2H,3C}$
3.27 bar
5h, 6h, 11h
9h
$\Delta T_{pinch,ORC}$
3.82 bar
3e
6e
8h
10h
1.85 bar
0.42 bar
7e--10e
5e
1h
4e
0.88 bar
13h
4h
$\Delta T_{pinch,HPE}$
2e
1e
$\Delta T_{E,sink}$